

OPPONG KWABENA ADUTWUM

CONTACT

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TECHNICAL SKILLS

Programming Languages

- Python
- C++

Data & Analytics

- Pandas / NumPy
- SQL / PostgreSQL
- Jupyter

Web & Backend

- FastAPI / Flask
- React Native / Django

Dev Tools

- Git / GitHub
- VS Code
- Linux (Kali)

Core Knowledge

- Machine Learning (fundamentals)
- Network Security
- Embedded Systems / IoT
- PID Control / Robotics

LANGUAGES

- English — Native
- French — Intermediate

CERTIFICATIONS

- Hashgraph Developer Course — Hedera (January 2026)
- Hedera Certified Foundation (April 2026)
- IPPF Competitor — Brewer Foundation & NYU (2024-2025)
- WASSCE General Science — Prempeh College (September 2025)
- [Deep Learning Spec.] — DeepLearning.AI
- [ML Specialization] — Stanford / Coursera

GITHUB HIGHLIGHTS

- [repo] — CNN crop disease classifier ★[X]
- [repo] — LLM-powered study assistant ★[X]
- [repo] — Sentiment analysis pipeline ★[X]
- [X] contributions | [X] PRs (12 mo.)

OPEN SOURCE

- [Hugging Face / PyTorch] — PR #[X]
- [scikit-learn / LangChain] — PR #[X]

RESEARCH & BLOG

- "[Title]" — [arXiv / Conference]
- "[Title]" — [Medium / Hashnode]

INTERESTS

- AI Safety & Alignment
- Computer Vision
- AI for Africa / Social Good
- Cybersecurity / CTFs

AI / MACHINE LEARNING ENGINEER

Deep Learning · Computer Vision · NLP · Robotics · Cybersecurity

PROFESSIONAL SUMMARY

AI/ML Engineering undergraduate at KNUST with hands-on experience in embedded intelligence, autonomous systems, and full-stack ML deployment. Regional Cybersecurity Champion (2024) and National Robotics Champion (Robofest Ghana 2025). Passionate about building intelligent systems—from PID-controlled robots to sensor-driven IoT. Seeking an AI/ML internship to design, train, and deploy models that solve real-world problems at scale.

EXPERIENCE

Machine Learning Intern — Zap Tech Technologies

Remote | May 2025

- Completed a 2-week intensive training program in the Machine Learning division, covering core ML concepts, algorithms, and workflows.
- Gained foundational exposure to supervised learning, data preprocessing, and model evaluation techniques.
- Collaborated with fellow interns on training exercises applying ML principles to real-world problem scenarios.

[AI/ML Engineering Intern] — [Company Name — e.g. Anthropic, Google DeepMind]

[City, Country] | [Mon YYYY – Mon YYYY]

- Trained and fine-tuned [model type] on [dataset], improving [metric] by [X]% over baseline.
- Built end-to-end ML pipeline: data ingestion → preprocessing → training → evaluation → deployment on [AWS/GCP].

[ML Research Assistant] — [KNUST AI Lab / Dept.]

Kumasi, Ghana | [Mon YYYY – Present]

- Researched [topic, e.g. NLP for low-resource African languages] under Prof. [Name].
- Implemented [model/algorithm] in PyTorch, achieving [X]% improvement on [benchmark].

EDUCATION

Kwame Nkrumah University of Science & Technology (KNUST)

B.Sc. Computer Science | Kumasi, Ghana | Expected 2029

Prempeh College

WASSCE — General Science | Kumasi, Ghana | Graduated 2025

PROJECTS

[Computer Vision Project — e.g. Crop Disease Detector]

[PyTorch · CNN/ResNet · FastAPI · AWS] | [github.com/...] [live-demo]

- Trained [CNN/ResNet/YOLO] on [X]K images achieving [X]% accuracy on [dataset]. Solved [real-world problem].
- Built inference pipeline: image upload → preprocessing → prediction → confidence score in [X]ms.
- Deployed via FastAPI on [AWS/GCP], serving [X] predictions/day. THIS is what Anthropic/Google want to see.

[NLP / LLM Project — e.g. AI Study Assistant]

[Claude/OpenAI API · LangChain · RAG · Next.js] | [github.com/...] [app-url]

- Built RAG pipeline: document ingestion → embedding → vector search → LLM generation with [X]% relevance.
- Implemented prompt chaining and guardrails for safe, grounded responses. [X] queries/day.

[Predictive ML Project — e.g. Energy Demand Forecasting]

[Scikit-learn · XGBoost · Pandas · Streamlit] | [github.com/...] [live-demo]

- Built [regression/classification] model with [X]% accuracy. Feature engineering from [X] raw features.
- Deployed interactive dashboard on Streamlit for real-time predictions.

Autonomous Line-Following Robot

LEGO MINDSTORM EV3 · PID Control Algorithm | [Video: youtube.com/...]

- Designed PID control algorithm (a core ML precursor) for autonomous navigation; National Champion at Robofest Ghana 2025.
- Optimized sensor-motor feedback loops — directly applicable to reinforcement learning and robotics control.

Class Attendance Monitoring System

React Native · Django · PostgreSQL · Bluetooth | [github.com/...]

- Architected biometric attendance platform with real-time data pipeline reducing manual roll-call by ~80%.
- Built Django REST API handling concurrent Bluetooth fingerprint scans — experience in data ingestion at scale.

Solar-Powered IoT Watering System

Arduino · IR Sensor · C++ · Solar | [github.com/...]

- Designed sensor-driven embedded system with proximity-triggered actuation — edge computing for IoT.
- Engineered full circuit with solar charge controller, relay logic, and sensor fusion.

HACKATHONS & COMPETITIONS

National Cybersecurity Challenge — Regional Champion (2024)

- Competed in a knowledge-based quiz testing cybersecurity fundamentals, threat analysis, and security best practices.

Robofest Ghana — National Champion — Robotics (2025)

Built autonomous robots under timed conditions to complete designated tasks; designed and assembled within strict time limits.

ACES CodeFest 2026 — UI/UX Challenge — Competitor (2026)

- Redesigned the ACES KNUST website into a mobile-first, student-focused experience across 9 pages in a 2-week sprint.

◆ LEADERSHIP & VOLUNTEERING

Club Leader — Prempeh College Cybersecurity Club (2023–2025)

Participant — International Public Policy Forum (IPPF) (2024)

Volunteer — Solutions for Life Initiative Ghana (SFLIG)

◆ MEMBERSHIPS & COMMUNITIES

Member — AmaliTech Coding Club (2025–Present)

Member — Microsoft User Group Ghana — KNUST Chapter (2025–Present)

Member — Claude Builder Club — KNUST (2025–Present)

Member — Prempeh College Robotics Club (2023–2025)